Lexicographically Smallest String

You are given a string `s` and an array `arr[]` of integers where `arr[i]` represents the weight of the character at index `i` in the string (1-based indexing).

You are allowed to swap the characters in the string that have the same weight in the array. In other words, if `arr[i] = arr[j]`, you can swap `s[i]` and `s[j]`.

Your task is to return the lexicographically smallest string that can be formed by applying any number of such swaps.

Input Format:

- A string `s` of length `N` consisting of lowercase English letters.
- An integer array `arr[]` of length `N`, where each element represents the weight of the corresponding character at index `i` in the string `s`.

Output Format:

Return the lexicographically smallest string that can be formed by swapping characters with the same weight.

Constraints:

- $1 \le N \le 2000$
- $1 \le arr[i] \le 10^4$
- String s consists of only lowercase English letters.

Sample Input	Sample Output
xvrb	bvrx
[2, 1, 2, 2]	

Explanation

Choose ‘x’, ’r’, ’b’ as they are of the same weight and sort them. ‘v’ does not have anyone to swap with so it remains in the same position.

Minimum price

There are N stones lying in a line. The cost and type of the i^{th} stone is a_i unit(s) and i respectively. You are initially having zero stones and you wish to collect all the N types of stones, type 1, type 2, ..., type N.

You can perform the following operation multiple times (probably zero) to change the types of all the stones in one step:

- The stone of the type i will be changed to the type i+1. If i is N, then change its type to 1. ($1 \leq i \leq N$)

Applying this operation single time costs x unit(s).

Your task is to print the minimum price that you have to pay to get all the N types of stones in your collection.

Input format

- First line: Two integers N, x
- Next line: N space-separated integers representing the price of each stone

Output format

Print a single integer representing the minimum price to get all the N types of stones.

Input Constraints

- $1 < N \leq 2000$
- $0 < x \leq 10^9$
- $1 \leq a[i] \leq 10^9$

Sample Input

3 5
50 1 50

Sample Output

13

Explanation

Operation 1: Buy 2^{nd} stone (stone of type 2) for price 1.
Operation 2: Pay price 5, the sequence of stone types becomes (2,3,1).
Operation 3: Buy 2^{nd} stone (stone of type 3) for price 1.
Operation 4: Pay price 5, the sequence of stone types becomes (3,1,2).
Operation 5: Buy 2^{nd} stone (stone of type 1) for price 1.
Total price paid is 1+5+1+5+1 = 13. You can't collect all types of stones by paying less than this.